

OSBORNE V. LOPES

AI Engineer

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SUMMARY

AI and Data Engineer with 3+ years of experience building production RAG pipelines, agentic multi-step workflows, and semantic search systems using LangChain, LangGraph, Qdrant, and Azure AI Search, alongside a background in data engineering across Snowflake, Kafka, Azure ADF, and DBT. Skilled at taking AI systems from prototype to production, improving retrieval quality, reducing latency, and integrating observability, guardrails, and evaluation to ensure reliable outputs.

SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, PySpark
AI & Automation: RAG, Agentic AI, LangChain, LangGraph, Hugging Face, FastAPI, ChromaDB, Qdrant, Pinecone, LLM, n8n, Prompt Engineering, Claude API, Gemini API, Model Context Protocol (MCP), Tool Calling, Vector Search, Embeddings, Reranking
Machine Learning: Machine Learning, Predictive Analytics, NLP, Feature Engineering, Model Evaluation, Statistical Analysis
Data Engineering: ETL/ELT, Data Pipelines, Data Modeling, Dimensional Modeling, Star Schema, dbt, Apache Airflow, Delta Lake
Cloud & AI Platforms: Microsoft Azure, Azure AI Search, Azure Data Factory, Azure Cloud Services
Data Platforms: Snowflake, SQL Server, Data Warehousing, Data Lake Architecture
Backend & APIs: FastAPI, RESTful APIs, API Integration, Microservices
Analytics & BI: Power BI, DAX, Tableau, Looker, Amazon QuickSight, Data Visualization, KPI Reporting
DevOps & Automation: CI/CD, Control-M, Pipeline Automation, Monitoring, Git

EXPERIENCE

ServiceNow, MA | AI Engineer Feb 2026 – Present

- Deployed a fine-tuned LLM chatbot on Azure Cloud Services handling 8,500+ daily employee queries, achieving 94% intent accuracy and reducing IT support ticket volume by 37% within 3 months.
- Led AI product go-to-market (GTM) initiatives by partnering with Product, Engineering, IT, and business stakeholders to define adoption strategies, pilot use cases, user feedback loops, and production rollout plans, accelerating enterprise adoption of AI-powered knowledge solutions.
- Built a FastAPI-based microservice with real-time vector search integration using Azure AI Search, enabling sub-200ms Similarity searches across 50M+ embedded vectors for document recommendation.
- Designed agent orchestration logic to manage multi-step state and conversation memory across workflows, evaluating LLM cost-performance trade-offs across model sizes and implementing self-critique, guardrails, and grounding mechanisms that improved response reliability by 30%.

Atlas SP, New York, NY | AI Engineer Co-op Jan 2025 – Aug 2025

- Built and deployed a production RAG-based chatbot using Azure AI Search vector indexes, LangChain, and FastAPI, enabling natural language querying of live financial data for internal teams and reducing repetitive analyst requests by 30%
- Improved retrieval quality by implementing hybrid search with a reranking layer to reprioritize retrieved passages before generation, integrating LangSmith for observability and monitoring key metrics like token usage and latency
- Built ADF batch-based and Kafka streaming-based ingestion pipelines from Azure Blob Storage to Snowflake, transforming data from raw to presentation layers and applying business logic through Alteryx workflows for downstream Tableau analytics
- Collaborated with team members through code reviews and PR merges to maintain pipeline quality, while partnering with business stakeholders to define KPIs and requirements for downstream Tableau dashboarding

Zyphron Industries, India | Analytics Engineer Jul 2021 – Aug 2023

- Designed star-schema semantic models in Power BI with advanced DAX measures to track inventory, fulfillment, and supplier KPIs, earning the trust of logistics leadership across 5+ business teams.
- Analyzed customer delivery KPIs across 10+ enterprise clients using complex SQL window functions and incremental DBT models, applying A/B testing to reporting formats that reduced repeat customer complaints by 15%.
- Reduced inventory shortfalls by 20% by analyzing historical inventory and order data with SQL and Excel to identify peak-season demand patterns and anticipate potential stockouts.
- Automated ETL pipelines using Python (Pandas, NumPy, Scikit-Learn) to preprocess and validate 100k+ daily shipment, supplier, and fulfillment records, cutting manual prep time by 50% and enabling faster, cleaner data for downstream reporting.

EDUCATION

Master of Science in Information Systems Dec 2025

Northeastern University, USA

Bachelor of Technology in Information Technology Jun 2021

University of Mumbai, India

PROJECTS

RAG Job Assistant

- Built a production-grade RAG pipeline indexing 5,000+ job postings into ChromaDB using Hugging Face sentence-transformer embeddings (all-MiniLM-L6-v2), enabling semantic search over job descriptions.
- Architected a multi-agent retrieval system using LangGraph with ReAct-pattern agents for query decomposition, semantic search, and answer synthesis, integrating LangChain and the Groq API (Llama 3.3 70B) with prompt engineering to manage context windows, multi-turn interactions, and output quality evaluation.

NewsScope

- Built an industry-specific news aggregator that scrapes, processes, and surfaces real-time business news using Python (Beautiful Soup, Scrapy) and DBT.
- Designed an end-to-end ETL pipeline on GCP using Dataflow for processing, Pinecone for vector embeddings, and BigQuery as the data warehouse, enforcing data quality via Pydantic 2 and PyTest validations.
- Containerized and deployed the platform with Docker, exposing a Streamlit interface for browsing curated news by industry.